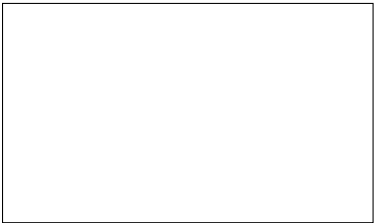


Graphical Abstract

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Kushal Umamahesh, Tamás Józsa



Highlights

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- Research highlights item 1
- Research highlights item 2
- Research highlights item 3

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ABSTRACT

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1. Introduction

Improving airfoil performance without active actuation remains an important objective in aerospace and energy systems. Conventional high-lift and flow-control devices can be effective, but they may introduce mechanical complexity, additional mass, maintenance requirements, and external geometric discontinuities. Passive flow-control concepts offer a complementary approach in which the pressure field surrounding an aerodynamic body supplies the energy required to modify the flow. Porous surfaces, permeable regions, discrete apertures, and internal passages all exploit this principle by permitting fluid transport between locations of unequal pressure. Their effectiveness, however, depends on where the openings are placed, how they are connected, and how strongly the resulting transpiration interacts with the external aerodynamic field.

Early investigations established that surface porosity can substantially alter the pressure loading of an airfoil, but also demonstrated that porosity is not inherently beneficial. Mineck and Hartwich examined full-chord porosity on a NACA 0012 airfoil in transonic flow and found that the porous surface generally reduced the lift-curve slope and increased drag at a specified normal-force coefficient [1]. Experiments by Aldheeb et al. similarly showed that increasing porosity reduced the lift-curve slope, while the drag response was non-monotonic, initially decreasing at low porosity before increasing as porosity became larger [2]. These results illustrate that unrestricted pressure equalisation can weaken the pressure difference responsible for lift and that porosity does not produce a universally favourable aerodynamic response. Useful passive flow control therefore requires selective permeability or controlled connectivity rather than indiscriminate porosity across the airfoil.

Reduced-order theories have provided important insight into the aerodynamic influence of distributed permeability. Hajian and Jaworski introduced a steady thin-airfoil formulation in which a local Darcy condition couples the

pressure jump across the airfoil to seepage through a chord-wise porosity distribution [3]. Luesutthiviboon et al. assessed a related thin-airfoil treatment for partially porous airfoils and identified reduced accuracy when the porous extent is small or the flow resistivity is low [4]. Baddoo et al. subsequently extended porous-airfoil theory to unsteady motion and gust response by incorporating resistance and effective-fluid-inertia terms into the porous boundary condition [5]. These models are computationally efficient and clarify the relationship between pressure difference, resistance, and aerodynamic loading. Nevertheless, thin-airfoil formulations neglect the finite thickness of the airfoil, and their porous conditions primarily describe local pressure–suction-side communication through a distributed or homogenised surface rather than arbitrary finite-length passages connecting selected, non-local surface locations.

Experiments and high-fidelity simulations also show that the internal structure and directionality of the permeable region are important. Hanna and Spedding studied a NACA 0012 wing containing discrete wall-normal holes and observed that a small degree of porosity modified the lift response around zero incidence while maintaining comparable lift-to-drag performance within the experimental uncertainty over much of the tested range [6]. For a NACA 0018 airfoil, Zhu et al. compared fully porous and suction-side porous trailing-edge arrangements and found distinct aerodynamic and acoustic responses, demonstrating the sensitivity of the outcome to porous placement and cross-surface pressure release [7]. Li et al. compared an isotropic porous medium with directionally constrained straight holes using large-eddy simulation and showed that aerodynamic behaviour depends not only on porous extent and porosity but also on the internal flow direction permitted by the structure [8]. Taken together, these studies show that a bulk porosity or permeability value alone is insufficient to describe the aerodynamic function of a designed internal architecture.

An alternative class of passive devices uses internal slots or bleed passages to transport fluid from a high-pressure region to a lower-pressure region. Acarer optimised passive slots for the DU12W262 airfoil and reported an improvement in peak lift-to-drag ratio, illustrating the potential of

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pressure-driven internal routing for aerodynamic enhancement [9]. Kestel et al. used passive bleeding through internal passages to alter the separated vortex structure over a non-slender delta wing, directly demonstrating flow transport driven by the natural pressure difference between connected surfaces [10]. More recently, Bahadi et al. parametrically investigated a curved passive internal slot in a NACA 4412 airfoil and reported improved high-incidence performance for selected inlet, taper, and curvature combinations [11]. These studies establish internal pressure-driven passages as prior art and therefore preclude broad claims of introducing passive surface connectivity. Most such investigations, however, consider a particular slot geometry using experiments or Reynolds-averaged Navier–Stokes calculations rather than a general reduced-order representation of multiple independently connected opening pairs.

Porous-airfoil design is increasingly being treated as an optimisation problem because the aerodynamic response depends on several interacting geometric and material variables. Zamani et al. applied a parameter-screening and optimisation procedure to a porous NACA 0015 airfoil for static-stall control, varying the porous-region dimensions, location, porosity, and cell size [12]. This work demonstrates the value of systematic design exploration, but the associated porous layer is described through distributed material parameters. For airfoils containing discrete passages, the design variables also include the identity and location of the connected surface openings, passage length, orientation, aperture, and the combination of same-surface and cross-surface routes. Exploring this larger topological design space exclusively with viscous computational fluid dynamics or experiments would be computationally expensive at the preliminary-design stage.

Panel methods provide an intermediate modelling level between analytical thin-airfoil theory and viscous computational fluid dynamics. The foundational work of Hess and Smith established the boundary-element representation of potential flow by distributing singularities over a body surface, enabling the pressure field around geometrically resolved bodies to be obtained without discretising the surrounding fluid domain [13]. The subsequent review by Hess documented the development, capabilities, and limitations of panel methods as efficient aerodynamic boundary-integral techniques [14], while Birk provided a modern and implementation-oriented treatment of the constant-source and dipole Hess–Smith formulation [15]. For a two-dimensional lifting airfoil, a source distribution can enforce the surface-normal boundary condition, while circulation and a trailing-edge Kutta condition provide the lifting solution. This structure is particularly suitable for the present problem because the impermeability condition can be replaced locally by a prescribed normal transpiration velocity at each passage opening. The resulting surface velocity gives the pressure coefficient directly, providing the endpoint pressure difference that drives the internal hydraulic calculation. Panel methods are also sufficiently inexpensive for repeated

solution across many passage layouts, sizes, and angles of attack.

Established viscous–inviscid methods further demonstrate how potential-flow panel solutions can form the aerodynamic core of more complete airfoil-analysis frameworks. Drela and Giles developed a strongly coupled viscous–inviscid method for transonic and low-Reynolds-number airfoils, combining an inviscid outer-flow solution with integral boundary-layer equations [16]. XFOIL subsequently provided a rapid airfoil-analysis and design system based on an inviscid linear-vorticity panel method coupled to integral boundary-layer and transition models [17]. These methods can predict viscous effects that are outside the present formulation, but their panel-based outer solutions support the use of potential flow for efficient surface-pressure calculation. In the present study, XFOIL is used only as a reference for the impermeable baseline; the proposed external-flow solver intentionally remains inviscid so that the pressure-driven aero–hydraulic coupling can be isolated and evaluated at low computational cost. This separation also makes the model suitable for preliminary screening, after which selected passage configurations can be examined using viscous computational fluid dynamics or experiments.

The reviewed literature therefore spans distributed Darcy-type surfaces, homogenised porous media, wall-normal holes, individual passive slots, and established potential-flow panel formulations. However, a modelling gap remains between local permeability boundary conditions and high-fidelity simulations of fixed internal geometries. In particular, limited attention has been given to a panel-based reduced-order framework that explicitly maps selected pairs of surface openings to finite-length passages, determines the pressure-driven flow through each passage, and feeds the resulting suction and blowing velocities back into the external aerodynamic solution. The systematic comparison of non-local same-surface passages, pressure-to-suction-surface passages, and their combined but hydraulically independent arrangement also remains underexplored. Addressing this gap would provide a computationally efficient basis for screening large numbers of candidate passage networks before higher-fidelity viscous simulation and experimental validation.

Accordingly, the present study develops a two-dimensional coupled aero–hydraulic framework for a NACA 0018 airfoil. The external flow is represented by a steady, incompressible, inviscid, and irrotational source–vortex panel method with a Kutta condition. Selected surface openings are connected by passive rectangular passages, and the pressure difference between the paired endpoints determines the internal flow through a hydraulic-resistance relation. The corresponding opening-normal velocities are iteratively coupled to the external panel solution. Three connectivity configurations are examined: four chordwise same-surface passages, four pressure-to-suction-surface passages, and a combined arrangement containing all eight passages as hydraulically independent connections. Passage heights of 8, 9, and 10 mm are assessed over angles of attack from -5° to 15° at a chord

Reynolds number of 5×10^5 , with comparisons against the impermeable airfoil and an XFOIL reference solution.

The principal contribution is the explicit reduced-order representation of prescribed, finite-length, pairwise surface connectivity within a coupled panel–hydraulic solver. The formulation permits different passage topologies to be evaluated within one consistent aerodynamic model and exposes the relationship between endpoint pressure difference, internal flow rate, surface transpiration, pressure distribution, lift, pitching moment, and pressure-based streamwise force. The method is intended for rapid preliminary assessment and topology screening. Because the external-flow model does not resolve boundary layers, skin-friction drag, separation, or stall, the results are interpreted as inviscid aerodynamic trends and a basis for selecting configurations for subsequent viscous computational and experimental validation.

2. Numerical methodology

2.1. Overall numerical framework

A two-dimensional steady aero–hydraulic framework was developed to investigate the aerodynamic behaviour of airfoils containing passive internal pores. The external flow is modelled using an incompressible, inviscid and irrotational source–vortex panel method, while the pressure-driven flow through each internal passage is evaluated using the Hagen–Poiseuille and Darcy–Weisbach relations, selected according to the Reynolds number of the flow within the circular pore. The aerodynamic and hydraulic models are coupled through the pressure and surface-normal velocity at the pore openings.

For each pore, the external aerodynamic solution provides the pressure at both ends of the pore. The resulting pressure difference drives flow through the internal passage from the higher-pressure end towards the lower-pressure end. This internal flow introduces suction at one end and blowing at the other, which are represented in the panel method through surface-normal transpiration velocities. The modified boundary condition then changes the external velocity and pressure fields, which in turn affect the pressure difference driving the pore flow. The external and internal flows are therefore solved as a coupled aero–hydraulic system.

The calculation begins from the corresponding impermeable (solid) airfoil solution, with the surface-normal transpiration velocity initially set to zero. The resulting surface pressures provide the initial pressure differences across the pores and hence the initial internal flow rates. The aerodynamic solution, pore pressure differences and transpiration velocities are then updated iteratively until a consistent coupled solution is obtained. This procedure is applied independently to each of the investigated pore heights, $H = 8, 9$ and 10 mm, and the converged aerodynamic and internal-flow quantities are retained for analysis.

The aerodynamic model assumes steady, incompressible, inviscid and irrotational flow. It therefore does not resolve boundary-layer development, skin-friction drag, flow

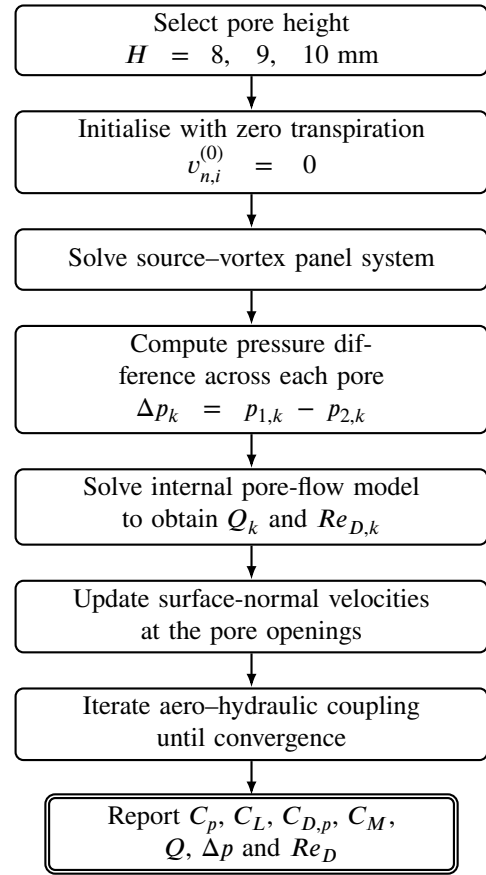


Figure 1: Coupled aero–hydraulic solution procedure applied to the investigated pore heights.

separation or stall. Consequently, the predicted drag coefficient represents only the pressure-drag contribution obtained from the potential-flow solution. The numerical procedure is summarised in Figure. 1.

2.2. Airfoil geometry and porous configurations

The baseline geometry is a NACA 0018 airfoil with chord c . The NACA four-digit coordinates are generated analytically and discretised using cosine spacing along the chord.

Each aerodynamic panel i is characterised by its length S_i , control point (x_i, y_i) , unit tangent vector

$$\mathbf{t}_i = (t_{x,i}, t_{y,i}), \quad (1)$$

and outward unit normal vector

$$\mathbf{n}_i = (n_{x,i}, n_{y,i}). \quad (2)$$

The porous structure is represented by passive circular pores formed as internal channels connecting selected locations on the airfoil surface. No prescribed mass-flow rate or external pumping mechanism is imposed. Instead, the magnitude and direction of the internal flow are determined entirely by the aerodynamic pressure difference between the two surface openings.

Three porous-airfoil configurations are considered:

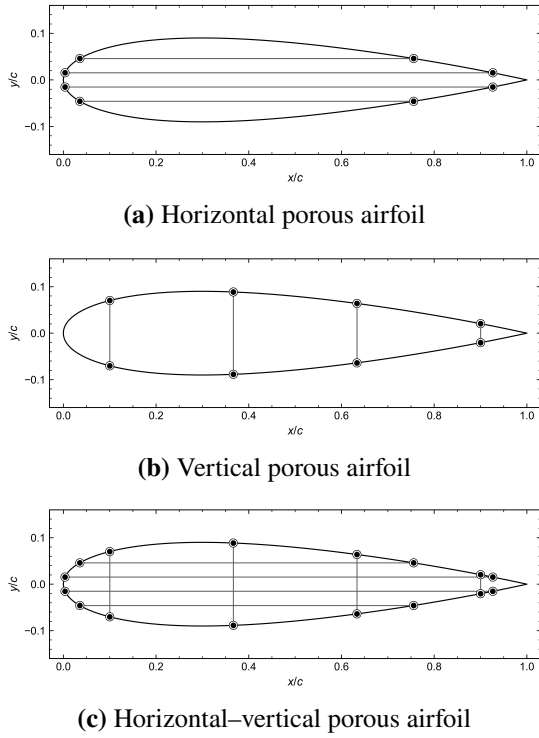


Figure 2: Porous-airfoil configurations considered in the present study: (a) horizontal porous airfoil, (b) vertical porous airfoil, and (c) horizontal-vertical porous airfoil.

1. **Horizontal porous airfoil:** four independent chord-wise pores are arranged with two located above and two below the chord line. Both ends of each pore are located on the same airfoil surface.
2. **Vertical porous airfoil:** four independent pores connect the lower and upper airfoil surfaces and are distributed between $x/c = 0.1$ and $x/c = 0.9$.
3. **Horizontal-vertical porous airfoil:** the horizontal and vertical pore arrangements are combined, resulting in eight internal pores. Although the pores may cross geometrically in the two-dimensional representation, they do not physically intersect or form internal junctions. Each pore remains hydraulically independent, and no flow is exchanged between the horizontal and vertical passages.

The three porous-airfoil configurations considered in the present study are illustrated in Figure. 2. The pore locations are mapped onto the upper and lower airfoil surfaces, allowing the prescribed pore positions to be specified independently of the discrete panel control points. This preserves the intended position and orientation of the internal passages without requiring them to coincide with specific panel locations.

The channel geometry is checked before the coupled calculation to ensure that the openings remain within the permitted chordwise region, do not overlap, fit within the local airfoil thickness and satisfy the specified minimum separation.

Each pore k is characterised by a height H_k and a centreline length L_k . The pore length is determined from the

geometry of the two corresponding surface openings and the internal passage connecting them.

2.3. External aerodynamic model

2.3.1. Potential-flow formulation

The external flow is assumed to be incompressible, inviscid and irrotational. The velocity field can therefore be expressed in terms of a scalar velocity potential,

$$\mathbf{V} = \nabla \Phi. \quad (3)$$

Application of mass conservation gives Laplace's equation,

$$\nabla^2 \Phi = 0. \quad (4)$$

The external-flow solution is obtained by superposing the uniform freestream with constant-strength source panels distributed along the airfoil surface and a single uniform vortex-sheet strength applied over the entire airfoil boundary. The source strength on each panel is determined by enforcing the normal-velocity boundary condition at the corresponding control point. The source panels represent the effect of the airfoil thickness, while the global vortex strength provides the circulation required to generate lift.

2.3.2. Source-vortex panel formulation

For an airfoil discretised into N panels, the aerodynamic unknowns are the N source strengths

$$\lambda = [\lambda_1, \lambda_2, \dots, \lambda_N]^T \quad (5)$$

and the global vortex strength γ .

The normal and tangential source influence coefficients are denoted by I_{ij} and J_{ij} , respectively, while K_{ij} and L_{ij} represent the corresponding vortex influence coefficients.

For an impermeable airfoil, the normal velocity at every control point is zero. To represent a pore opening, this condition is modified by prescribing a surface-normal transpiration velocity $v_{n,i}$,

$$\mathbf{V}_i \cdot \mathbf{n}_i = v_{n,i}. \quad (6)$$

For an impermeable airfoil, the surface-normal velocity is zero, such that

$$v_{n,i} = 0. \quad (7)$$

Following discretisation, the normal boundary condition for panel i becomes

$$\pi \lambda_i + \sum_{\substack{j=1 \\ j \neq i}}^N I_{ij} \lambda_j - \gamma \sum_{j=1}^N K_{ij} = -2\pi V_\infty \cos \beta_i + 2\pi v_{n,i}, \quad (8)$$

where β_i denotes the angle between the local panel normal and the freestream direction.

The trailing-edge Kutta condition is imposed by requiring the tangential velocities on the first and final panels to be equal in magnitude and opposite in direction,

$$V_{t,1} + V_{t,N} = 0. \quad (9)$$

The N normal boundary conditions together with the Kutta condition produce an $(N + 1) \times (N + 1)$ linear system,

$$\begin{bmatrix} \mathbf{A}_\lambda & \mathbf{a}_\gamma \\ \mathbf{a}_K^\top & a_{K,\gamma} \end{bmatrix} \begin{bmatrix} \lambda \\ \gamma \end{bmatrix} = \begin{bmatrix} \mathbf{b}(\mathbf{v}_n) \\ b_K \end{bmatrix}. \quad (10)$$

The influence matrix depends only on the discretised airfoil geometry. It is therefore assembled and LU-factorised once for each geometry, while the right-hand side is updated according to the freestream condition and surface-normal transpiration velocity.

2.3.3. Surface velocity and pressure

Following solution of the panel system, the tangential velocity on panel i is evaluated as

$$V_{t,i} = V_\infty \sin \beta_i + \frac{1}{2\pi} \sum_{j=1}^N \lambda_j J_{ij} + \frac{\gamma}{2} - \frac{\gamma}{2\pi} \sum_{j=1}^N L_{ij}. \quad (11)$$

At a porous opening, the local surface velocity contains both tangential and normal components. The velocity magnitude is therefore

$$V_i^2 = V_{t,i}^2 + v_{n,i}^2. \quad (12)$$

The pressure coefficient is obtained from the steady incompressible Bernoulli equation,

$$C_{p,i} = 1 - \frac{V_{t,i}^2 + v_{n,i}^2}{V_\infty^2}. \quad (13)$$

The corresponding absolute pressure is

$$p_i = p_\infty + \frac{1}{2} \rho_\infty V_\infty^2 C_{p,i}. \quad (14)$$

2.4. Internal pore-flow model

The internal flow through each pore is modelled as pressure-driven flow between parallel walls. Each pore k is characterised by a pore height H_k and a centreline length L_k . Consistent with the two-dimensional formulation, the flow is evaluated per unit span, with a hydraulic diameter

$$D_{h,k} = 2H_k. \quad (15)$$

The flow is driven by the pressure difference between the two ends of the pore,

$$\Delta p_k = p_{1,k} - p_{2,k}. \quad (16)$$

A positive or negative value of Δp_k therefore determines the direction of flow through the pore. The mean internal velocity is related to the volumetric flow rate per unit span, q'_k , by

$$\bar{U}_k = \frac{q'_k}{H_k}. \quad (17)$$

The corresponding pore Reynolds number is

$$Re_{h,k} = \frac{\rho_\infty |\bar{U}_k| D_{h,k}}{\mu_\infty}. \quad (18)$$

The pressure loss through the pore is calculated using the Darcy–Weisbach relation,

$$|\Delta p_k| = f_k \frac{L_k}{D_{h,k}} \frac{\rho_\infty \bar{U}_k^2}{2}, \quad (19)$$

where f_k is the Darcy friction factor. For laminar flow between parallel walls, $f_k = 96/Re_{h,k}$. At higher Reynolds numbers, the Blasius correlation is used, with a smooth transition between the two relations over $2000 < Re_{h,k} < 3000$.

For a given pressure difference, the resulting pressure-loss relation is solved to determine the internal flow rate. The calculated flow is then used to obtain the surface-normal transpiration velocity imposed on the external aerodynamic solution.

2.5. Aero-hydraulic coupling

The external aerodynamic and internal pore-flow models are coupled through the surface pressure and the normal velocity imposed at the two ends of each pore.

The pressure at each end of a pore is obtained from the aerodynamic surface-pressure distribution. For a prescribed pore height, the corresponding surface panels are identified from the aerodynamic discretisation, and their pressure values are used to determine the pressure associated with that end. For end m of pore k ,

$$p_{m,k} = \frac{\sum_{i \in \mathcal{P}_{m,k}} p_i \ell_i}{\sum_{i \in \mathcal{P}_{m,k}} \ell_i}, \quad m = 1, 2, \quad (20)$$

where $\mathcal{P}_{m,k}$ represents the panels covered by the pore end and ℓ_i is the corresponding overlap length.

The pressure difference across each pore is then

$$\Delta p_k = p_{1,k} - p_{2,k}, \quad (21)$$

which provides the driving pressure for the internal pore-flow model. The resulting flow rate per unit span, q'_k , gives the mean velocity through the pore as

$$\bar{U}_k = \frac{q'_k}{H_k}. \quad (22)$$

This velocity is introduced into the aerodynamic model as a surface-normal transpiration velocity. With the outward surface normal defined as positive, flow entering the pore is assigned a negative normal velocity, while flow leaving the pore is assigned a positive normal velocity,

$$v_{n,k}^{\text{in}} = -\bar{U}_k, \quad v_{n,k}^{\text{out}} = +\bar{U}_k. \quad (23)$$

When a pore end extends across several panels, the transpiration condition is applied over the corresponding surface region rather than at a single panel control point.

The imposed normal velocity alters the external flow and therefore the surface-pressure distribution. This changes the

pressure difference driving the internal pore flow, which in turn updates the transpiration velocity. The aerodynamic and pore-flow solutions are therefore updated iteratively until the pressure difference and internal flow are consistent with one another. This procedure is repeated independently for each pore height considered in the study, and all reported cases reached a converged coupled solution.

2.6. Aerodynamic coefficients

The aerodynamic forces are obtained by integrating the pressure contribution over the discretised airfoil surface. The contribution of panel i to the global Cartesian force coefficients is

$$dC_{x,i} = -C_{p,i} n_{x,i} \frac{S_i}{c}, \quad (24)$$

$$dC_{y,i} = -C_{p,i} n_{y,i} \frac{S_i}{c}. \quad (25)$$

The total Cartesian force coefficients are

$$C_x = \sum_{i=1}^N dC_{x,i}, \quad C_y = \sum_{i=1}^N dC_{y,i}. \quad (26)$$

Rotation into the freestream-aligned coordinate system gives the lift and pressure-drag coefficients,

$$C_L = C_y \cos \alpha - C_x \sin \alpha, \quad (27)$$

$$C_{D,p} = C_x \cos \alpha + C_y \sin \alpha. \quad (28)$$

The pitching-moment coefficient is evaluated about the quarter-chord reference location,

$$(x_{\text{ref}}, y_{\text{ref}}) = (0.25c, 0), \quad (29)$$

using

$$C_M = - \sum_{i=1}^N \frac{(x_i - x_{\text{ref}})dC_{y,i} - (y_i - y_{\text{ref}})dC_{x,i}}{c}. \quad (30)$$

For diagnostic comparison, an additional lift coefficient is calculated from the solved circulation using the Kutta–Joukowski relationship,

$$C_{L,\Gamma} = \frac{2\gamma \sum_{i=1}^N S_i}{V_\infty c}. \quad (31)$$

2.7. Numerical setup

The aerodynamic performance of each porous configuration is compared with that of the corresponding impermeable airfoil under the same geometry, panel discretisation and operating conditions. An inviscid XFOIL solution is also used as an additional reference to compare the impermeable-airfoil pressure distribution and its variation with angle of attack.

Unless otherwise stated, all calculations are performed for a NACA 0018 airfoil using the numerical and freestream conditions summarised in Table 1.

Table 1

Numerical and operating conditions used in the present study.

Parameter	Symbol	Value
Airfoil	–	NACA 0018
Chord length	c	1 m
Number of panels	N	1000
Fluid density	ρ_∞	1.225 kg m ⁻³
Dynamic viscosity	μ_∞	1.8 × 10 ⁻⁵ Pa s
Freestream pressure	p_∞	101325 Pa
Chord Reynolds number	Re_c	5.0 × 10 ⁵
Principal angle of attack	α	4°
Angle-of-attack range	α	–5° to 15°
Angle-of-attack increment	$\Delta\alpha$	1°
Pore height	H	8, 9, 10 mm

The freestream velocity is determined from the prescribed chord Reynolds number as

$$V_\infty = \frac{Re_c \mu_\infty}{\rho_\infty c}. \quad (32)$$

Each angle of attack is solved independently, and the solution from the preceding angle is not used to initialise the subsequent case.

3. Results and discussion

The aerodynamic behaviour of the three porous-airfoil configurations was investigated over an angle-of-attack range of –5° to 15°. Three pore heights, $H = 8, 9$ and 10 mm, were considered for the horizontal, vertical and horizontal–vertical porous airfoils. The corresponding impermeable (solid) airfoil panel-method solution and the inviscid XFOIL prediction are included as reference cases.

The effect of the pores is first examined through the lift and moment coefficients over the angle-of-attack range. The observed aerodynamic trends are then interpreted using the corresponding pressure and velocity field differences. A detailed flow-field comparison is presented at $\alpha = 10^\circ$ for a pore height of $H = 9$ mm.

3.1. Panel method validation against XFOIL

The impermeable source–vortex panel-method solution is compared with the corresponding inviscid XFOIL prediction in Figures. 3 and 4. Both methods show similar trends in lift and pitching moment over the investigated angle-of-attack range.

This agreement provides a useful reference for assessing the aerodynamic effects introduced by the pores. Small differences between the panel-method and XFOIL predictions are expected because the two approaches use different numerical formulations, although both represent inviscid external flow.

The impermeable panel-method solution is therefore used as the primary reference when evaluating the porous-airfoil configurations. This ensures that the porous and solid

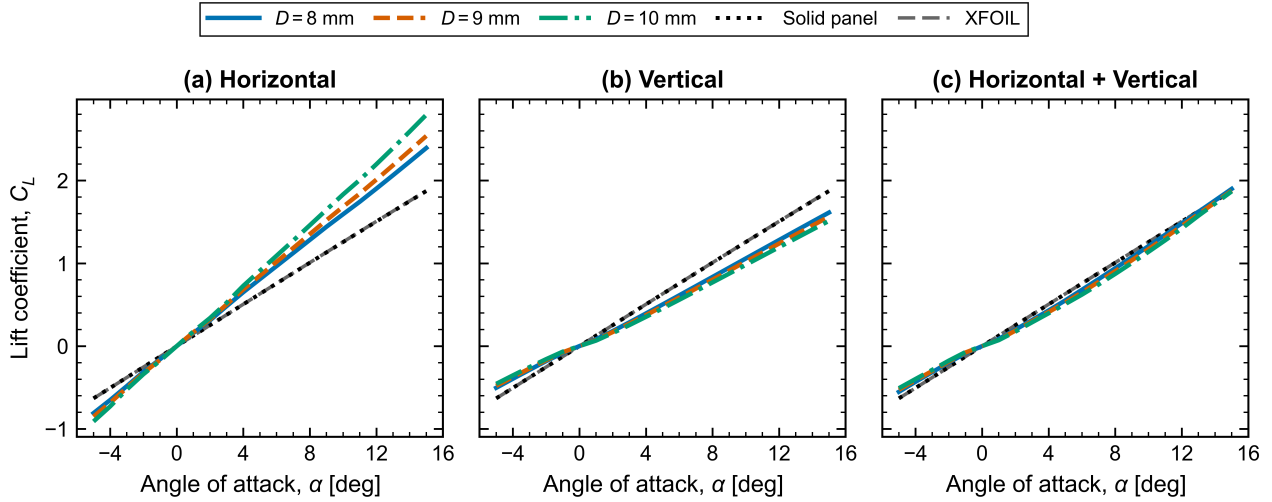


Figure 3: Lift coefficient as a function of angle of attack for (a) the horizontal porous airfoil, (b) the vertical porous airfoil, and (c) the horizontal–vertical porous airfoil. Results are shown for pore heights $H = 8, 9$ and 10 mm together with the impermeable panel-method and inviscid XFOIL reference solutions.

airfoils are compared using the same geometry, panel discretisation, aerodynamic formulation and operating conditions.

3.2. Effect of pores on lift

Figure 3 compares the variation of lift coefficient with angle of attack for the three porous-airfoil configurations and pore heights. The horizontal porous airfoil produces the largest change in lift. At positive angles of attack, its lift coefficient is higher than that of the impermeable reference, and the difference becomes more pronounced as the angle of attack increases. The effect also strengthens with pore height, with $H = 10$ mm producing the highest lift at larger positive angles of attack. At negative angles of attack, increasing the pore height similarly increases the magnitude of the negative lift. This nearly symmetric lift response about the zero-lift condition is consistent with the symmetric NACA 0018 geometry.

For the horizontal configuration, both ends of each pore are located on the same airfoil surface. The streamwise variation in surface pressure therefore creates a pressure difference between the two ends and drives internal flow from the locally higher-pressure region towards the lower-pressure region. This produces suction at one end and blowing at the other, changing the local surface-normal velocity and redistributing the external pressure and velocity fields along the airfoil surface. At positive angle of attack, the different pressure distributions over the suction and pressure surfaces lead to a net redistribution of aerodynamic loading that increases lift.

The stronger dependence on pore height is consistent with the reduction in hydraulic resistance as the pore becomes larger. For a given pressure difference, a larger pore height permits a greater internal flow and therefore a stronger interaction between the pore flow and the external aerodynamic field.

The vertical porous airfoil shows a distinctly different behaviour. At positive angles of attack, its lift coefficient is generally lower than that of the impermeable airfoil, while the three pore-height cases remain relatively close over most of the investigated range.

In this configuration, each pore connects the pressure surface to the suction surface. The resulting pressure difference drives internal flow between the two surfaces and partially reduces the pressure difference responsible for lift. This pressure equalisation is consistent with the lower lift coefficient obtained for the vertical porous airfoil. The small separation between the pore-height cases also indicates that this configuration is less sensitive to pore height than the horizontal arrangement.

The horizontal–vertical porous airfoil remains comparatively close to the impermeable reference. At positive angles of attack, its lift coefficient is generally slightly lower than that of the solid airfoil, and the three pore-height cases remain closely grouped.

This behaviour reflects the simultaneous influence of the horizontal and vertical pore arrangements. The horizontal pores redistribute flow between different chordwise locations on the same surface, whereas the vertical pores transfer flow between the pressure and suction surfaces. The lift-enhancing redistribution associated with the horizontal pores is therefore partly offset by the pressure equalisation introduced by the vertical pores.

The horizontal and vertical pores remain physically separate and hydraulically independent. However, their aerodynamic effects interact through the external pressure field, since changes produced by one set of pores alter the surface pressures that drive the other set. Consequently, the response of the horizontal–vertical porous airfoil is not a simple superposition of the two individual configurations.

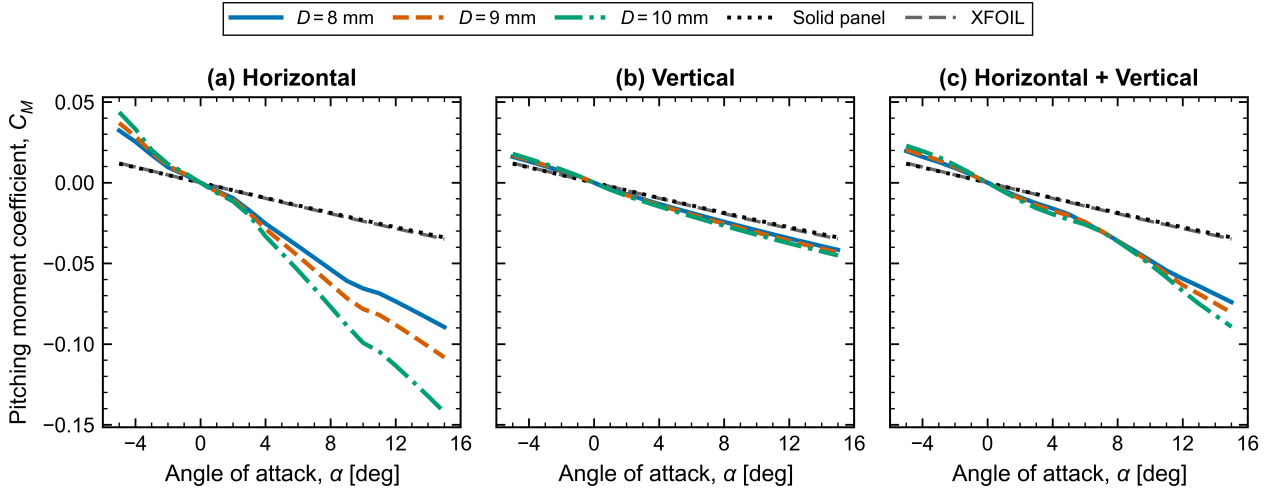


Figure 4: Pitching-moment coefficient about the quarter chord as a function of angle of attack for (a) the horizontal porous airfoil, (b) the vertical porous airfoil, and (c) the horizontal-vertical porous airfoil. Results are shown for pore heights $H = 8, 9$ and 10 mm together with the impermeable panel-method and inviscid XFOIL reference solutions.

Overall, the results show that the aerodynamic effect of the pores is governed mainly by their orientation, placement and the pressure difference between the connected surface locations. Pore height also influences the response, but its effect is strongest for the horizontal configuration and considerably weaker for the vertical and horizontal-vertical arrangements.

3.3. Effect of pores on pitching moment

The corresponding moment coefficients are presented in Figure. 4. The horizontal porous airfoil produces the largest increase from the impermeable-airfoil response. As the positive angle of attack increases, the pitching-moment coefficient becomes progressively more negative. The magnitude of this change also increases with pore height, with $H = 10$ mm producing the largest negative pitching moment at higher positive angles of attack.

This behaviour is associated with the chordwise redistribution of surface pressure caused by the horizontal pores. Since the two ends of each pore are located at different chordwise positions along a given airfoil surface, the pressure changes induced by the internal flow act at different distances from the quarter-chord reference point. These local pressure variations therefore contribute differently to the pitching moment. Increasing the pore height strengthens the internal flow and the resulting pressure redistribution, leading to a larger change in the pitching-moment coefficient.

The increase in lift obtained with the horizontal porous airfoil is therefore accompanied by a larger negative pitching moment. This represents an important aerodynamic trade-off, since the configuration that provides the greatest lift enhancement also produces the largest change in longitudinal loading.

The vertical porous airfoil produces considerably smaller changes in moment, and the three pore-height cases remain

closely grouped over most of the investigated angle-of-attack range. This behaviour is consistent with the weaker dependence on pore height observed in the lift results.

For the vertical configuration, the two ends of each pore are positioned on the upper and lower surfaces at approximately the same chordwise location. The resulting pressure redistribution therefore occurs mainly between the pressure and suction surfaces rather than between widely separated positions along the chord. The corresponding change in the chordwise pressure distribution is smaller than for the horizontal pores, resulting in a weaker pitching-moment response.

The horizontal-vertical porous airfoil exhibits behaviour between that of the individual horizontal and vertical configurations. The pitching moment becomes increasingly negative with positive angle of attack, although the departure from the impermeable-airfoil solution remains smaller than for the horizontal porous airfoil. The influence of pore height also becomes more noticeable at higher positive angles of attack.

In this configuration, the horizontal pores redistribute loading between different chordwise positions, while the vertical pores mainly redistribute pressure between the upper and lower surfaces. The two pore arrangements remain physically separate and hydraulically independent, but their aerodynamic effects interact through the external pressure field. The resulting pitching-moment variation therefore reflects the combined influence of these two pressure-redistribution mechanisms.

Overall, the pitching-moment results show that the effect of the pores depends not only on the extent of the pressure redistribution, but also on where it occurs along the chord. The horizontal porous airfoil produces the largest change in moment because the associated pressure variations occur at widely separated chordwise locations, creating a stronger moment about the quarter-chord reference point. In contrast,

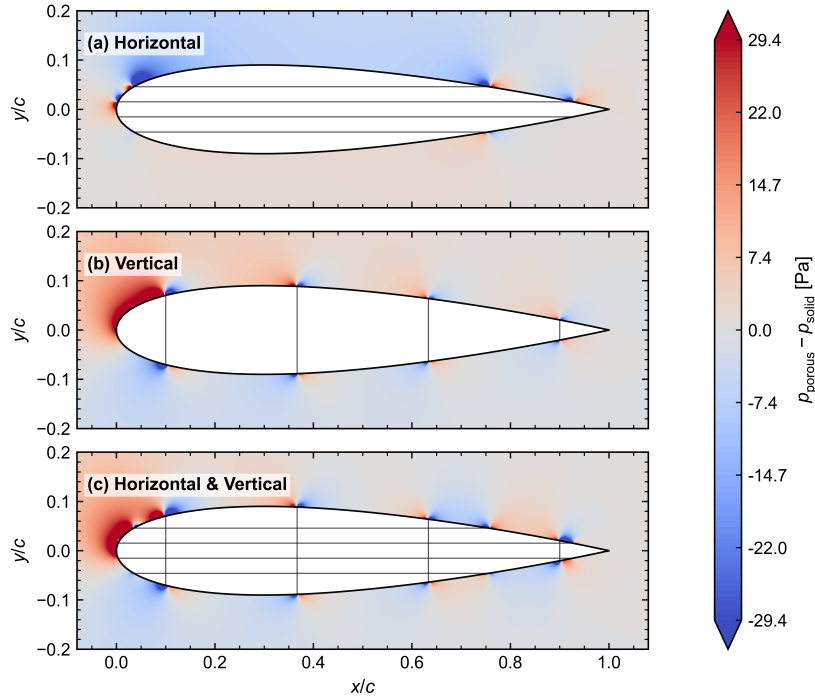


Figure 5: Pressure-field difference between the porous and impermeable airfoils at $\alpha = 10^\circ$ and $H = 9$ mm for (a) the horizontal porous airfoil, (b) the vertical porous airfoil, and (c) the horizontal–vertical porous airfoil. Positive values indicate an increase in pressure relative to the impermeable airfoil, while negative values indicate a reduction.

the vertical porous airfoil produces a smaller change because its pressure redistribution occurs mainly between the upper and lower surfaces at similar chordwise positions.

3.4. Flow field characteristics of porous airfoils

To examine how the pores alter the external flow, the pressure and velocity fields of the porous airfoils are compared with those of the corresponding impermeable airfoil at $\alpha = 10^\circ$ and $H = 9$ mm.

The comparison is presented using pressure- and velocity-difference contours rather than the absolute flow fields. The absolute contours are dominated by the overall aerodynamic behaviour of the airfoil and therefore make the local influence of the pores more difficult to identify. Subtracting the impermeable-airfoil solution isolates the changes introduced by the pore flow, allowing the location, direction and relative strength of the resulting pressure and velocity variations to be examined more clearly.

3.4.1. Pressure field characteristics

Figure 5 presents the pressure difference between the porous and impermeable airfoils,

$$\Delta p = p_{\text{porous}} - p_{\text{solid}}, \quad (33)$$

for the three pore arrangements.

The pressure difference is shown instead of the absolute pressure field to isolate the changes introduced by the pores. In the absolute solution, the pressure distribution is dominated by the overall airfoil loading and the strong

pressure gradients near the leading edge. Subtracting the impermeable-airfoil solution makes the local changes associated with the pore flow easier to identify.

The strongest pressure differences occur near the pore ends, where the internal flow directly influences the surrounding external flow. In several locations, neighbouring positive and negative values of Δp form a dipole-like pattern. This is associated with suction at one end of a pore and blowing at the other, producing local regions of reduced and increased pressure relative to the impermeable airfoil.

The effect of these pressure changes on lift depends on where they occur on the airfoil. At positive angle of attack, a reduction in pressure on the upper suction surface strengthens the suction and contributes to higher lift, whereas an increase in pressure has the opposite effect. On the lower pressure surface, an increase in pressure contributes positively to lift, while a pressure reduction tends to decrease it.

For the horizontal porous airfoil, the pressure differences are mainly concentrated around the chordwise pore ends. Because both ends of each pore lie on the same airfoil surface, the internal flow redistributes pressure between different chordwise locations. The combined pressure changes on the upper and lower surfaces increase the pressure difference across the airfoil, consistent with the higher lift coefficient observed for this configuration.

The vertical porous airfoil produces a different pattern because each pore connects the pressure surface to the suction surface. Flow is therefore driven from the higher-pressure region towards the lower-pressure region, producing local pressure changes at both pore ends. At the same

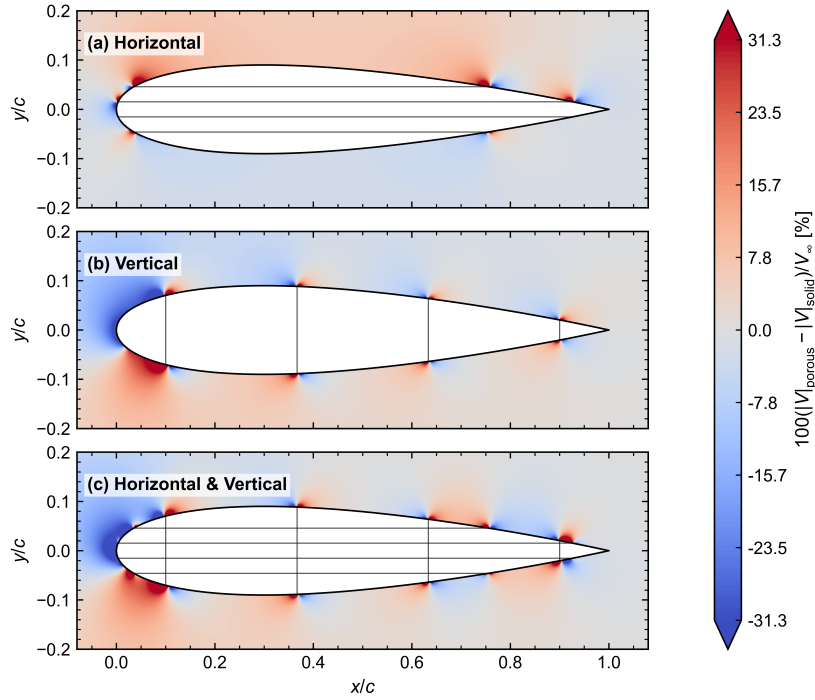


Figure 6: Percentage difference in velocity magnitude between the porous and impermeable airfoils at $\alpha = 10^\circ$ and $H = 9$ mm for (a) the horizontal porous airfoil, (b) the vertical porous airfoil, and (c) the horizontal–vertical porous airfoil. Positive values indicate an increase in velocity magnitude, while negative values indicate a reduction relative to the impermeable airfoil.

time, this cross-surface flow partially reduces the pressure difference between the two surfaces, which is consistent with the lower lift coefficient obtained for the vertical configuration.

The horizontal–vertical porous airfoil contains features associated with both pore arrangements. The horizontal pores redistribute pressure along the airfoil surfaces, while the vertical pores transfer flow between the pressure and suction surfaces. Although the two pore systems remain physically separate and hydraulically independent, their aerodynamic effects interact through the external pressure field. Consequently, the pressure equalisation associated with the vertical pores partly counteracts the pressure redistribution produced by the horizontal pores, giving an overall aerodynamic behaviour closer to that of the impermeable airfoil.

3.4.2. Velocity field characteristics

The velocity field is presented as the percentage difference relative to the impermeable-airfoil solution,

$$\Delta|V|_{\%} = \frac{|V|_{\text{porous}} - |V|_{\text{solid}}}{|V|_{\text{solid}}} \times 100. \quad (34)$$

This comparison isolates the changes introduced by the pores from the dominant background flow around the airfoil. The largest velocity differences occur close to the pore ends, where the imposed surface-normal flow locally perturbs the external velocity field. Adjacent regions of acceleration and deceleration are visible around several pores and correspond closely to the pressure patterns shown in Figure. 5.

The velocity and pressure fields show the expected inverse relationship described by Bernoulli's equation. Regions where the flow accelerates are generally associated with a reduction in static pressure, whereas regions of lower velocity correspond to higher static pressure. The velocity contours therefore help explain how the pore-induced changes in the local flow field lead to the observed pressure redistribution and, consequently, changes in the aerodynamic forces and pitching moment.

For the horizontal porous airfoil, alternating regions of acceleration and deceleration are concentrated around the chordwise pore ends. Since both ends of each pore lie on the same airfoil surface, the internal flow redistributes the external velocity between different chordwise locations. The combined changes on the suction and pressure surfaces strengthen the pressure difference across the airfoil, consistent with the increase in lift and the larger change in pitching moment observed for this configuration.

The vertical porous airfoil shows a different velocity pattern. Flow between the pressure and suction surfaces changes the external velocity on both sides of the airfoil and reduces part of the velocity contrast between them. This behaviour is consistent with the partial pressure equalisation discussed previously and the corresponding reduction in lift. A broader region of velocity difference is also visible near the leading edge, where the external flow experiences strong pressure and velocity gradients.

The horizontal–vertical porous airfoil contains features associated with both pore arrangements. The horizontal

pores produce local chordwise acceleration and deceleration, while the vertical pores redistribute flow between the pressure and suction surfaces. Although the two pore arrangements remain physically separate and hydraulically independent, their aerodynamic effects interact through the external flow field. The resulting velocity distribution therefore reflects the combined influence of both mechanisms rather than a simple superposition of the individual configurations.

Overall, the velocity contours support the trends observed in the pressure field and aerodynamic coefficients. The horizontal pores primarily redistribute the flow along the airfoil surfaces, whereas the vertical pores transfer flow between the pressure and suction surfaces. These different flow mechanisms contribute to the contrasting lift and pitching-moment behaviour of the three porous-airfoil configurations.

3.5. Overall comparison of porous configurations

The combined aerodynamic and flow-field results show that the response of the porous airfoil depends strongly on the orientation and placement of the pores, while the influence of pore height varies between the different configurations.

Among the three configurations, the horizontal porous airfoil produces the strongest aerodynamic effect. At positive angles of attack, it increases the lift coefficient relative to the impermeable airfoil, with the enhancement becoming more pronounced as the pore height increases. This increase in lift is accompanied by a more negative pitching moment, indicating that the pores also redistribute the aerodynamic loading along the chord.

The vertical porous airfoil shows a different behaviour. The pores connect the higher-pressure pressure surface to the lower-pressure suction surface, allowing flow to pass between the two sides of the airfoil. This partially reduces the pressure difference responsible for lift and results in a lower lift coefficient than the impermeable reference at positive angles of attack. The relatively small separation between the pore-height cases also indicates a weaker sensitivity to pore height, together with smaller changes in pitching moment.

The horizontal-vertical porous airfoil combines both pore arrangements. The horizontal and vertical pores remain physically separate and hydraulically independent, but their aerodynamic effects interact through the external pressure and velocity fields. The pressure redistribution associated with the vertical pores partly offsets the lift-enhancing effect of the horizontal pores, causing the overall aerodynamic response to remain closer to that of the impermeable airfoil.

The pressure- and velocity-difference fields support these trends. The largest local changes occur near the pore ends, where the internal flow alters the surrounding external flow. The horizontal pores mainly redistribute the flow between different chordwise locations on the same surface, whereas the vertical pores transfer flow between the pressure and suction surfaces. The resulting aerodynamic response therefore depends not only on pore height, but also on the

surface locations connected by each pore and the pressure difference established between them.

Overall, the horizontal porous airfoil provides the greatest potential for lift enhancement among the configurations considered. However, the associated increase in the magnitude of the pitching moment shows that the performance of a porous-airfoil design should not be assessed from lift alone. Pore orientation, placement and height should therefore be considered together with their effects on both aerodynamic forces and moments.

4. Conclusion

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